

KATHMANDU UNIVERSITY SCHOOL OF MEDICAL SCIENCES
END SEMESTER EXAMINATION
10th Batch 1st Year BDS

Subject: Anatomy

Date: 6th May 2022
Time- 2 hrs. 30 minutes

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

Section B [25 Marks]

- 1) Enlist the type of synovial joint with one example of each. [6]
- 2) Mention the histological differences between skeletal muscle and cardiac muscle. [5]
- 3) Draw and label the histological diagram of spleen. [5]
- 4) Write **short notes** on: [3×3=9]
 - a. Neurulation
 - b. Blood supply of typical long bone
 - c. Gross features of palatine tonsil.

Section C [25 Marks]

- 5) Mention the derivatives of branchial apparatus. [4+2=6]
- 6) Write down the boundaries and contents of axilla. [5]
- 7) Mention the formation and branches of lumbosacral plexus. [5]
- 8) Write **short notes** on: [3×3=9]
 - a. Long saphenous vein
 - b. Lacrimal apparatus
 - c. Anatomical snuffs box

KATHMANDU UNIVERSITY SCHOOL OF MEDICAL SCIENCES

END SEMESTER EXAMINATION

10th Batch 1st Year BDS

Subject: Physiology

Date: 8th May 2022

Time- 2 hrs. 30 minutes

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

Section B [25 Marks]

- 1) Describe the ionic basis of action potential in a nerve fibre with its well labeled diagram. [5]
- 2) Compare and contrast primary active transport and secondary active transport along with an example of each. [6]
- 3) Explain the basis of classification of human beings into Rh system. Add a note on its importance in blood transfusion. [2+3=5]
- 4) Write **short notes** on: [3 × 3=9]
 - a) Negative Feedback mechanism
 - b) Functions of plasma proteins
 - c) Neutrophils

Section C [25 Marks]

- 5) Define synapse. Write the steps of synaptic transmission with diagram. [1+5=6]
- 6) Enumerate the functional difference between sympathetic and parasympathetic nervous system with regard to their role in heart and urinary bladder. [2.5+2.5=5]
- 7) Classify leucocytes. Write down the stages of erythropoiesis. [2+3=5]
- 8) Write **short notes** on: [3 × 3=9]
 - a) Role of platelets in hemostasis
 - b) Landsteiner's law
 - c) Refractory period

KATHMANDU UNIVERSITY
May/June 2017

Level : BDS

Subject: Anatomy

Exam : BDS II (old course)

Paper : II

Time : 3 hour

Marks : 70

Date : May 31, 2017

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
Write brief, relevant and legible answers
Illustrate your answer with well labeled diagrams whenever required
All questions are compulsory

SECTION "A" [35 marks]

- a. Mention the extent and normal constrictions present in the oesophagus. [2+3=5]
- b. Describe briefly arterial supply of stomach. [5]
- c. Draw a labeled histological diagram of thick skin. (253) Hand drawn [5]
- d. Describe briefly the relations and arterial supply of pituitary gland. [3+2=5]
- e. Write short notes on:
a. Features of autosomal dominant mode of inheritance [2×5=10]
b. Sites of porto-caval anastomosis
- f. Draw a labeled histological diagram of fallopian tube. [5]

SECTION "B" [35 marks]

6. Enlist the muscles of mastication with their innervation. [3+2=5]
7. Describe the boundaries, contents and applied anatomy of carotid triangle. [2+2+1=5]
8. Draw a labeled diagram showing any five functional cortical area of cerebrum. [5]
9. Describe the histological structure of ureter with well labeled diagram. [3+2=5]
10. Write short notes on:
a. Scalp
b. Trigon of urinary bladder [3×5=15]
c. Circle of Willis

KATHMANDU UNIVERSITY
July/August 2018

Level : BDS

Subject: Anatomy

Exam : BDS I (new course)

Paper : I

Time : 2 hr. 30 minutes

Marks : 50

Date : July 6, 2018

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"

[25 marks]

1. Define epithelium and classify it with examples. [5]
2. Write down the boundaries, contents and applied anatomy of axilla. [5]
3. Describe the types of joint with their examples. [6]
4. Write short note on: [3×3=9]
 - a. Elbow joint
 - b. Thick skin
 - c. Arches of foot

SECTION "C"

[25 marks]

5. Describe briefly temporomandibular joint under flowing headings: [5]
 - a. Articulating bones
 - b. Ligaments and relations
6. Describe the development of inter-atrial septum with its anomalies. [5]
7. Write down the boundaries, contents and applied anatomy of carotid triangle. [6]
8. Write short note on: [3×3=9]
 - a. Scalp
 - b. Typical intercostal nerve
 - c. Pericardium

KATHIMANDU UNIVERSITY
December 2018/January 2019

Level : BDS

Subject: Anatomy

Exam : BDS I (new course)

Paper : I

Time : 2 hr. 30 minutes

Marks : 50

Date : December 31, 2018

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"

[25 marks]

1. Describe the development of placenta with its anomalies. [3+2=5]
2. Describe the formation, branches and applied anatomy of brachial plexus. [2+2+1=5]
3. Write down the differences between hyaline, elastic and white-fibro cartilages. [2+2+2=6]
4. Write short notes on: [3×3=9]
 - a. Hip joint
 - b. Axillary artery
 - c. Sympathetic nervous system

SECTION "C"

[25 marks]

5. Describe the boundaries, contents and applied of infratemporal fossa. [2+2+1=5]
6. Describe the development of lung with its anomalies. [3+2=5]
7. Describe the boundaries, contents and applied of superior mediastinum. [2+2+2=6]
8. Write short notes on: [3×3=9]
 - a. Anterior triangle of neck
 - b. Intercostal space
 - c. Maxillary air sinus

July/August 2019

1 : BDS

Subject: Anatomy

n : BDS I (new course)

Paper : I

e : 2 hour 30 minutes

Marks : 50

: : August 1; 2019

INSTRUCTIONS TO THE CANDIDATE

Answer each section in a separate answer book

Write brief, relevant and legible answers

Illustrate your answer with well labeled diagrams whenever required

All questions are compulsory

SECTION "B"

[25 marks]

Describe the histological structures of spleen with well labeled diagram. [6]

Describe the stages of intra cartilaginous ossification. [5]

Describe structure of a neuron with a labeled diagram. [5]

Write short notes on:

[3×3=9]

a. Autonomic ganglia

b. Carpal tunnel syndrome

c. Movements of knee joint

SECTION "C"

[25 marks]

Describe the characteristic features and components of a bronchopulmonary segment with a labeled diagram. [6]

Describe briefly internal features of right atrium and mention the structures opening into it. [5]

Describe formation, relations, tributaries and development of superior vena cava. [5]

Write short notes on:

[3×3=9]

a. Sternocleidomastoid muscle

b. Carotid triangle

c. Sternal angle

KATHMANDU UNIVERSITY
February/March 2020

Level : BDS

Exam : BDS I

Time : 2 hour 30 minutes

Date : February 26, 2020

Subject: Anatomy

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Define ossification. Describe the different stages of intra cartilaginous ossification. [1+5 = 6]
2. Describe the histological structures of lymph node with well labeled diagram. [5]
3. Explain the boundary, contents and applied anatomy of popliteal fossa. [5]
4. Write short notes on: [3×3=9]
 - a. Brachial plexus
 - b. Thick skin
 - c. Spermiogenesis

SECTION "C"
[25 marks]

5. Name muscles of mastication with their origin, insertion, action and nerve supply. [6]
6. Describe the skeletal framework of thoracic cage. [5]
7. Describe the blood supply of heart with clinical applications. [5]
8. Write short notes on: [3×3=9]
 - a. Lacrimal apparatus
 - b. Paranasal air sinuses
 - c. Apex of lung

KATHMANDU UNIVERSITY
August/ September 2021

Level : BDS

Subject: Anatomy

Exam : BDS I (new course)

Paper : I

Time : 2 hr. 30 minutes

Marks : 50

Date : August 25, 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Mention the microscopic features of lymph node with the help of labeled diagram. 337
[3+3=6]
2. Mention the formation and extent of clavipectoral fascia and enumerate the structures piercing it. [2+2+1=5]
3. Define joint. Classify the synovial joint with examples. [1+3+1=5]
4. Write short notes on: [3×3=9]
 - a. Gastrulation.
 - b. Popliteal fossa.
 - c. Histological difference between thick and thin skin.

SECTION "C"
[25 marks]

5. Describe the development of interatrial septum with a well labeled diagram and mention its congenital anomalies. [4+2=6]
6. Mention the boundaries and contents of posterior triangle of the neck with the help of labeled diagram. [2+3=5]
7. Describe gross features and development of diaphragm. [3+2=5]
8. Write short notes on: [3×3=9]
 - a. Tributaries of coronary sinus.
 - b. Facial vein.
 - c. Trachea.

KATHMANDU UNIVERSITY
October/November 2021

Level : BDS

Exam : BDS I (new course)

Time : 2 hr. 30 minutes

Date : October 22, 2021

Subject: Anatomy

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Describe the histological structure of smooth muscle with a well labeled diagram. [6]
2. Describe briefly about the splenic circulation. [5]
3. Describe the formation, branches and applied aspects of brachial plexus. [1+2+2=5]
4. Write short notes on: [3×3=9]
 - a. Movement of temporomandibular joint.
 - b. Histology of thin skin.
 - c. Scalp.

SECTION "C"
[25 marks]

5. Describe external features and arterial supply of the heart. [3+3=6]
6. Describe boundaries, contents and applied aspects of typical intercostal spaces. [1+2+2=5]
7. Explain parts, recesses & applied anatomy of pleura of the lung. [1+2+2=5]
8. Write short notes on: [3×3=9]
 - a. Azygous vein.
 - b. Cartilages of larynx.
 - c. External carotid artery.

KATHMANDU UNIVERSITY
University Examination
February/March 2022

Level : BDS

Subject: Anatomy

Exam : BDS I

Paper : I

Time : 2 hr. 30 minutes

Marks : 50

Date : March 2, 2022

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Classify bones and write about arterial supply of long bone. [4+2=6]
2. What is Oogenesis? Explain its mechanism adding a note on "ovulation". [1+2+2=5]
3. Write about the formation of Brachial Plexus with suitable diagram. Mention any two applied anatomy of Brachial Plexus. [4+1=5]
4. Write short notes on: [3×3=9]
 - a. Femoral sheath
 - b. Monozygotic and dizygotic twin
 - c. Neurulation

SECTION "C"
[25 marks]

5. Explain the muscles of mastication with emphasis on attachment, nerve supply and action. [2+2+2=6]
6. Discuss the interior of right atrium with a well labelled diagram. [3+2=5]
7. Draw a labeled histological diagram of trachea. [5]
8. Write short notes on: [3×3=9]
 - a. Maxillary air sinus
 - b. Left coronary artery
 - c. Carotid triangle

Kathmandu University School of Medical Sciences
BDS Program
Internal Examination, June 2022

Subject: Oral Biology

Time: 2 hours 30 Minutes

Full marks: 50

Level: BDS I (New course)

Date: 6th June, 2022

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labeled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. What is the composition of enamel? Discuss Amelogenesis in detail. [1+5=6]
2. Describe in detail the morphology of permanent maxillary left canine with well labeled diagram. [5]
3. Discuss the functions of pulp. Add a note on Plexus of Raschkow. [3+2=5]
4. Write short notes on: [3x3=9]
 - a. Line angles and point angles
 - b. Hypocalcified structures of enamel
 - c. Elevated landmarks of tooth

Section "C" [25 marks]

5. Enumerate the morphological stages of tooth development. Discuss about transient structures. [1+4=5]
6. Classify salivary glands. Differentiate between serous and mucous acini. [2+3=5]
7. Enlist the layers of dental apparatus. Add a note on reciprocal induction and describe the process of formation of root. [1+2+3=6]
8. Write short notes on: [3x3=9]
 - a. Tooth numbering system
 - b. Occlusal aspect of maxillary first premolar
 - c. Formation and secretion of saliva

KATHMANDU UNIVERSITY SCHOOL OF MEDICAL SCIENCES
END SEMESTER EXAMINATION
10th Batch 1st Year BDS

Subject: Biochemistry

Date: 10th May 2022
Time- 2 hrs. 30 minutes

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

Section B [5 × 5=25 Marks]

- 1) Describe and classify carbohydrates with examples
- 2) Classify amino acid and write down the functions of proteins
- 3) Describe structural organization of protein with examples.
- 4) Define and classify lipids. List out the importance of phospholipids.
- 5) What are the differences between competitive and non-competitive inhibition? Give two examples for competitive inhibition.

Section C [5 × 5=25 Marks]

- 6) Describe the reactions of HMP Shunt pathway and Write about significance.
- 7) Write down the pathway of hem synthesis.
- 8) Explain with reactions, breakdown of glucose to pyruvate.
- 9) Describe iron metabolism and its regulation with the help of a diagram.
- 10) What is porphyria? Classify different types of porphyrias. Give an account of acute intermittent porphyria.

KATHMANDU UNIVERSITY SCHOOL OF MEDICAL SCIENCES
END SEMESTER EXAMINATION
10th Batch 1st Year BDS

Subject: Microbiology

Date: 15th May 2022
Time- 2 hrs. 30 minutes

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

Section B [25 Marks]

- 1) Define sterilization and disinfection. Write down the principle and uses of autoclave. [3+3=6]
- 2) Classify different types of culture media with examples. [5]
- 3) Discuss different method of antibiotic susceptibility testing. How do you interpret Kirby-Bauer disk diffusion method of antibiotic susceptibility testing? [5]
- 4) Write short notes on: [3×3=9]
 - a. Louis Pasteur
 - b. Bacterial Growth Curve
 - c. Normal flora

Section C [25 Marks]

- 1) Describe the lab diagnosis of malaria. [6]
- 2) Describe briefly the life cycle of *Wucheria bancrofti*. Add a note on its diagnosis. [2+3=5]
- 3) Describe the diagnosis of *Histoplasma capsulatum*. [5]
- 4) Write short notes on: [3×3=9]
 - a. Hypersensitivity reaction
 - b. Oncogenic viruses
 - c. Leishmaniasis

KATHMANDU UNIVERSITY SCHOOL OF MEDICAL SCIENCES

END SEMESTER EXAMINATION

10th Batch 1st Year BDS

Subject: Pharmacology

Date: 20th May 2022

Time- 2 hrs. 30 minutes

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

Section B [25 Marks]

- 1) What is first pass metabolism? Write in short about factors affecting the absorption of drugs.
[2+4=6]
[5×2=10]
- 2) Write briefly on:
 - a. Prodrug
 - b. Plasma half life
 - c. Disadvantages of oral route of drug administration.
 - d. Essential drug concept.
 - e. Rational drug therapy
[3×3=9]
- 3) Write **short notes** on:
 - a. Potency and efficacy with illustration.
 - b. List of different phase I and phase II metabolic reactions.
 - c. Streptokinase

Section C [25 Marks]

- 4) Classify anticoagulant drugs. Write down the mechanism of action warfarin. Mention its important therapeutic uses.
[3+2+1=6]
- 5) Write the classification of adrenergic drugs on the basis of their therapeutic effects with at least one example of each. Write down the therapeutic uses Prazosin.
[3 + 2=5]
- 6) Explain the mechanism of action of Chloroquine. Write down its important therapeutic uses and adverse effects.
[3+1+1=5]
[3×3=9]
- 7) Write **short notes** on:
 - a. Organophosphorous poisoning and its management.
 - b. Therapeutic uses Atropine
 - c. Therapeutic uses of Propranolol

KATHMANDU UNIVERSITY SCHOOL OF MEDICAL SCIENCES
END SEMESTER EXAMINATION
10th Batch 1st Year BDS

Subject: Pathology

Date: 18th May 2022
Time- 2 hrs. 30 minutes

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

Section B [25 Marks]

- 1) Describe the pathogenesis of septic shock, stages of shock and write down the morphological changes in shock. [3+2+1=6]
- 2) Describe vascular and cellular events of inflammation. Write factors affecting wound healing. [3+2=5]
- 3) Write mechanism of Apoptosis. Describe modes of spread of malignant tumor. [3+2=5]
- 4) Write **short notes** on:
 - a. Pathological calcification
 - b. Paraneoplastic syndrome
 - c. Carcinogenesis

Section C [25 Marks]

- 5) Define anemia. Write laboratory diagnosis of Iron deficiency anemia. [1+5=6]
- 6) Write laboratory diagnosis on: [2×5=10]
 - a. Acute myelogenous leukemia
 - b. Multiple myeloma.
- 7) Write **short notes** on: [3×3=9]
 - a. Morphology of Tuberculous lymphadenitis
 - b. Morphology of Hodgkins lymphoma
 - c. Etiopathogenesis of Disseminated intravascular coagulation